

Matrix Spaces; Rank 1; Small World Graphs

Matrices can be vectors too. A covariance matrix lives in one such space, and a single-factor market is a **rank one** matrix.

$$\dim S = n(n+1)/2$$

$$A = uv^T$$

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Anything you can add and scale

All 3×3 matrices form a space M of dimension 9 (basis: 9 matrices with a single 1).

Subspace	Dimension
Symmetric S	6
Upper triangular U	6
Diagonal $D = S \cap U$	3

$\dim S + \dim U = \dim(S + U) + \dim(S \cap U): 6 + 6 = 9 + 3.$

Covariance lives in S

A covariance matrix is symmetric. For n assets you need

$$\frac{n(n+1)}{2} \text{ numbers}$$

- 2 assets: 3 (two variances, one covariance)
- 3 assets: 6
- 50 stocks: 1,275

Diagonal covariance = uncorrelated assets.

A column times a row

$$\begin{bmatrix} 1 \\ 1.2 \\ 0.5 \end{bmatrix} \begin{bmatrix} -10 & 5 & 20 \end{bmatrix} = \begin{bmatrix} -10 & 5 & 20 \\ -12 & 6 & 24 \\ -5 & 2.5 & 10 \end{bmatrix}$$

Betas $\times$ market returns: every row a multiple of one row. **One factor drives everything $\Rightarrow$ rank 1.**

Rank $r = r$ rank-one pieces

Add a small-cap factor to midcap:

$+(0, 1, 0)^T (4, -2, 6)$. Midcap's row becomes $(-8, 4, 30)$, no longer a multiple of Nifty's.

Two factors $\Rightarrow$ rank 2.

Rank 1 + rank 1 is usually rank 2, so rank-one matrices are **not** a subspace.

THE MARKET MODEL

$$\mathbf{Cov} = \sigma^2 \boldsymbol{\beta} \boldsymbol{\beta}^T + \mathbf{D}$$

$$\underbrace{\begin{bmatrix} 400 & 480 & 0 \\ 480 & 576 & 0 \\ 0 & 0 & 0 \end{bmatrix}}_{400 \boldsymbol{\beta} \boldsymbol{\beta}^T \text{ (rank 1)}} + \underbrace{\begin{bmatrix} 0 & 0 & 0 \\ 0 & 324 & 0 \\ 0 & 0 & 25 \end{bmatrix}}_D$$

$$= \begin{bmatrix} 400 & 480 & 0 \\ 480 & 900 & 0 \\ 0 & 0 & 25 \end{bmatrix}$$

50 STOCKS

Full covariance: 1,275 numbers. One factor: $50 + 1 + 50 = 101$.

How far apart are two brokers?

Nodes = brokers, edges = direct trading links.

Distance = fewest links between two nodes.

The adjacency matrix A has a 1 for each link;
 $(A^2)_{ij}$ counts **two-step** paths.

In finance networks, distance is how many hops a shock takes to spread.

Key takeaways

1. Matrices form vector spaces: 3×3 has dim 9.
2. Symmetric: $n(n + 1)/2$, where covariance lives.
3. Rank one = uv^T : a single-factor model.
4. Rank $r = r$ rank-one pieces = r factors.
5. Graphs: distance, adjacency, A^2 .

QUANT FROM FIRST PRINCIPLES

The full interactive lesson is free

A factor builder, a rank one lab, and worked hedging and options examples:

learn.krisarj.com

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